



Pulse Dispatch

Enterprise SMS Gateway & Multi-Tenant Management Platform

Official Operational Manual for Platform Administrators (Superusers) and Business Clients (Tenants). Covers web administration, contact imports, campaign scheduling, and companion Android SIM hardware setup.

Version 1.0.0 • Production Release

Table of Contents

1. Platform Architecture & Multi-Tenant Isolation

2. Superuser (Administrator) Operations Guide

2.1 Logging in as Superadmin

2.2 Managing Client Accounts (/users)

2.3 Creating New Client Users

2.4 Disabling & Enabling Client Accounts

2.5 Resetting Client Passwords

2.6 Inspecting Client Hardware & Live Telemetry

3. Client User Guide (Business Tenants)

3.1 Web Dashboard Overview (/dashboard)

3.2 Importing Contacts via Excel Spreadsheet (/imports)

3.3 Managing Customer Directory (/customers)

3.4 Scheduling Events (/events)

3.5 Creating Message Templates (/templates)

3.6 Launching SMS Campaigns (/campaigns)

3.7 Real-time Broadcast Audit Logs (/logs)

4. Android Companion App Setup & SIM Gateway

4.1 Installing the Production APK

4.2 Configuring the Server URL

4.3 Logging In with Client Credentials

4.4 Permissions & Automatic SIM Dispatch

4.5 Activity Stream & Support Tickets

5. Production Deployment & Server Publishing

6. Troubleshooting & FAQs

1. Platform Architecture & Multi-Tenant Isolation

Pulse Dispatch turns standard Android smartphones into **private, low-cost SMS broadcast gateways**. The platform connects a web application management suite with native Android hardware dispatch.

Strict Multi-Tenant Data Isolation Rules

- **Client Users (CLIENT)**: Strictly see and manage their own customer records, scheduled events, broadcast campaigns, message templates, delivery logs, and companion Android phones.
- **Android Phone Gateway**: An Android phone logged in with a client account **strictly pulls and dispatches SMS jobs for that specific client's campaigns**.
- **Superuser (SUPERADMIN)**: Has global platform oversight, access to user provisioning at /users , 1-click account access control, and hardware telemetry auditing.

Feature	Client Tenant (CLIENT)	Superadmin (SUPERADMIN)
Data Scope	Strictly isolated to their own records	Global access across all client tenants
Users Menu (/users)	Hidden & inaccessible	Full user creation, disable/enable, reset
SMS SIM Dispatch	Companion phone sends only their SMS jobs	Can audit all connected devices & queues
Account Control	Self profile updates	1-Click Enable/Disable any user account

2. Superuser (Administrator) Operations Guide

2.1 Logging in as Superadmin

- 1 Navigate to the web portal login page (e.g. `http://localhost:3000/login` or your public domain).
- 2 Enter default Superadmin credentials:
Email: `pulak@example.com`
Password: `admin12345`
- 3 Upon sign in, the exclusive **Users & Clients** navigation tab will appear in the sidebar.

2.2 Managing Client Accounts (/users)

The **Users & Clients** dashboard provides complete control over tenant provisioning and system security:

- **Live Metric Cards:** Real-time counts for Total Users, Active Clients, Disabled Accounts, and Registered Android Phones.
- **Multi-parameter Filter:** Search by client name, email, company, or phone; filter by Role and Active status.
- **User Directory Table:** Displays company name, contact info, role badge, status pill, connected phone count, and action buttons.

2.3 Creating New Client Users

- 1 Click + **Create Client User** at the top right of /users .
- 2 Fill in client details:
 - **Full Name:** Primary contact name (e.g. David Miller).
 - **Email (Login ID):** Client username (e.g. david@apexretail.com).
 - **Initial Password:** Secure password for client (e.g. ApexPass2026!).
 - **Company Name:** Organization name (e.g. Apex Retail Pty Ltd).
 - **Phone & Address:** Contact number and city location.
 - **Role:** Select Client (Standard Tenant) .
- 3 Click **Create User Account**. The client can now log in to the web dashboard and Android app immediately.

2.4 Disabling & Enabling Client Accounts

⚡ 1-Click Instant Access Control

If a client's subscription expires or needs suspension, click the **Disable** button on their row in `/users`. The account turns **DISABLED (RED)**, and all web and mobile login or dispatch requests are immediately blocked with HTTP 403 Forbidden. Click **Enable** at any time to instantly restore access.

2.5 Resetting Client Passwords

Click the **Key icon (Reset Password)** on any client row, enter a new password, and click Confirm.

2.6 Inspecting Client Hardware & Activity Logs

Click the **Eye icon (View Activity)** on a client's row to open the **Client Activity Drawer**. This reveals:

- **Linked Android Hardware:** Device model (e.g. `Galaxy A55`), SIM carrier (`Telstra AU`), SIM phone number, battery percentage, and live status.
- **Live Audit Stream:** Timestamped logs of SMS dispatches, device logins, and support inquiries.

3. Client User Guide (Business Tenants)

3.1 Web Dashboard Overview (`/dashboard`)

The client dashboard is customized for broadcast operations:

- **Ready to Send:** Real-time count of customer contacts with active SMS consent.
- **Active Campaigns:** Live progress bars showing sent vs. pending messages.
- **Linked Phone:** Hardware telemetry and battery percentage of your companion Android phone.

3.2 Importing Contacts via Excel Spreadsheet (`/imports`)

- 1 Navigate to **Imports** in the sidebar.
- 2 Drag and drop your `.xlsx`, `.xls`, or `.csv` spreadsheet. Columns for Name, Mobile, Email, Company, City, and Tags are automatically mapped.
- 3 Enter your customized broadcast text using merge tags (e.g. `Hi [Name], your order is ready!`).
- 4 Click **Import & Queue Campaign**. Contacts are verified, deduplicated, and stored in your private database directory.

3.3 Managing Customer Directory (/customers)

View, search, and manage customer records with real-time SMS and Email consent status pills. Click + **Add Customer** to add individual clients manually.

3.4 Scheduling Events (/events)

Create event records with Date, Venue, Target Audience, and customized SMS message templates. Click **Launch Campaign** on any event card to trigger the broadcast.

3.5 Creating Message Templates (/templates)

Store reusable SMS and Email messages with dynamic merge tags like `{{name}}` , `{{company}}` , and `{{venue}}` .

3.6 Launching SMS Campaigns (/campaigns)

- 1 Click + **Create Campaign** on /campaigns .
- 2 Select channel **SMS (Android SIM)**, pick an optional linked Event, and set message text.
- 3 Click **Launch & Queue SMS**. Messages appear in the live queue for your phone to dispatch.

4. Android Companion App Setup & SIM Gateway

The companion Android application (**Pulse Sender**) turns your phone into a private carrier SIM SMS gateway.

4.1 Installing the Production APK

- 1 Obtain the production APK file: `android-app/app/build/outputs/apk/release/app-release.apk` .
- 2 Transfer the APK to any Android phone (Android 10.0+ / API 29+). Tap the file to install (allow "Install unknown apps" if prompted).

4.2 Configuring the Server URL

- 1 Open **Pulse Sender** on the phone.
- 2 Tap  **API Gateway Server** below the password field.
- 3 Enter your published server URL:
 - Production Domain: `https://your-domain.com`
 - Local Testing: `http://192.168.1.50:4000`

4.3 Logging In with Client Credentials

Enter the client **Email** and **Password** created by the Superadmin and tap **Sign In**. The app authenticates and displays your **CLIENT** badge and company name.

4.4 Permissions & Automatic SIM Dispatch

- **Send SMS Permission:** Required to send messages via the phone SIM card.
- **Notification Permission:** Keeps the background service active for automated dispatching.
- **Automatic Sending:** When campaigns are queued from the web app, your phone automatically pulls pending jobs and dispatches them with real-time status reporting.

5. Production Deployment & Server Publishing

Backend & Database (Docker Deployment)

```
git clone https://github.com/pulok529/SMSSendingAPP.git
cd SMSSendingAPP
docker compose up -d --build
```

The Express API runs on port `4000` and PostgreSQL on port `5432` . Configure an Nginx reverse proxy with SSL certificate (`https://api.yourdomain.com` → `http://localhost:4000`).

Web Application (Next.js)

```
npm ci
npm run build
npm run start
```

Proxy port `3000` with SSL (`https://yourdomain.com` → `http://localhost:3000`).

6. Troubleshooting & FAQs

Issue	Possible Cause	Solution
"Cannot reach server" on Mobile App	Incorrect Server URL or phone has no internet	Tap "API Gateway Server" on login screen and verify URL (e.g. <code>https://api.yourdomain.com</code>). Check Wi-Fi/data.
"Your account has been disabled"	Superadmin deactivated the client account	Contact administrator to reactivate your account in <code>/users</code> .
SMS jobs stay in "PENDING"	Phone is offline or background service paused	Open Pulse Sender app, check SIM signal, and ensure "Auto-Fetch Queue" is enabled in Settings.
Android kills background service	Aggressive battery saver on phone	Android Settings → Apps → Pulse Sender → Battery → Select Unrestricted .